Definition 7.4.1. A functor $F : \mathcal{C} \to \mathcal{D}$ is an **equivalence of categories** if there exists $F' : \mathcal{D} \to \mathcal{C}$ such that $F \circ F'$ and $F' \circ F$ are isomorphic to the identity functors.

In this situation, F' is said to be **a quasi-inverse** to F.

In particular, the above categories $\mathcal{C}_1$ and $\mathcal{C}_2$ are equivalent (check it!).

Also, the category **FSet** of finite sets is equivalent to the category whose objects are nonnegative integers, and morphisms are given by

$$\mathrm{Hom}(m, n) = \mathrm{Maps}(\{1, \ldots, m\}, \{1, \ldots, n\}).$$

Are these categories isomorphic? The answer to this question depends on whether you believe that there is only one finite set with a given number of elements, or that there are many of those. It seems better to think that there are many (without asking "how many"), so that isomorphic sets need not be literally equal, but this is really a matter of choice. In any case, this is not really a reasonable question; the answer to this question is irrelevant for any practical purpose, and thinking about it will give you nothing but a headache.

7.5. Representable functors

A fundamental notion in category theory is that of a **representable functor**. Namely, let $\mathcal{C}$ be a (locally small) category, and let $F : \mathcal{C} \to \mathbf{Sets}$ be a functor. We say that F is **representable** if there exists an object $X \in \mathcal{C}$ such that F is isomorphic to the functor $\mathrm{Hom}(X, ?)$. More precisely, if we are given such an object X, together with an isomorphism $\xi : F \cong \mathrm{Hom}(X, ?)$, we say that the functor F is **represented by** X (using ξ).

In a similar way, one can talk about representable functors from $\mathcal{C}^{\mathrm{op}}$ to **Sets**. Namely, one calls such a functor representable if it is of the form $\mathrm{Hom}(?, X)$ for some object $X \in \mathcal{C}$, up to an isomorphism.

Not every functor is representable, but if a representing object X exists, then it is unique. Namely, we have the following lemma.

Lemma 7.5.1 (The Yoneda Lemma)**.** *If a functor F is represented by an object X, then X is unique up to a unique isomorphism. I.e., if X, Y are two objects in $\mathcal{C}$, then for any isomorphism of functors*